

(Section - B) Theory

* Part 1: Short Answer

Q1 What is Virtual DOM?

Ans Virtual DOM is a lightweight copy of the Real DOM used by React. When changes happen, React updates the Virtual DOM first, compares it with the previous version, and then updates only the changed part in the Real DOM.

Q2 What are React Hooks?

Ans React Hooks are functions that allow you to use state and other React features in the functional components.

Ex: use state, use effect, use context

Q3 Difference between the state and props.

Ans Difference :-

State	Props
1, State Managed inside Component.	Props Passed from Parent Component.
2, It ^{It} can be changed.	Props are Read-only.
3, It used for Dynamic data	It is used to Pass Data

Q4 What is Lifting state up?

Ans Lifting state up means moving state from a child component to a common parent component so multiple components can share and use the same data. It is used for better data flow and synchronization.

Q5 What is Key Prop?

Ans The Key Prop is a Unique Identifier used in lists to

help React Identify which items have changed, added or Moved. It avoids Rendering issues

* Part 2:-

Q6 Explain useEffect Dependency Array

Ans The dependency array in useEffect controls when the effect runs.

Ex! - `useEffect(() => {
 // Code
}, [dependencies]);`

- `[]` :- runs one
- `[value]` :- runs when value change
- no array :- runs on every Render.

It helps to optimize performance and avoid unnecessary executions.

Q7 What is Prop Drilling and How to avoid it?

Ans Prop drilling is when data is passed through multiple components (Parent $\rightarrow$ Child $\rightarrow$ grandchild) even if the intermediate components don't need it.

* To avoid Props drilling we use, useContext API. And can also use state management tools (Redux, Zustand).

Q8 Controlled vs Uncontrolled components

Ans In Controlled components, ~~form~~ data is handled by React state. React controls the Input value. Easy validation and control.

In Uncontrolled components, ~~form~~ data is handled by the

DOM itself using refs. In uncontrolled components, No direct React control.

Q.9 What is React.memo?

Ans React.memo is a higher-order component that prevents unnecessary re-renders.

ex: - `export default React.memo(MyComponent);`

It re-renders only when props change.

Q.10 What is useCallback Vs useMemo?

Ans	useCallback	useMemo
1.	Memoizes a function	Memoizes a value
2.	It Returns a function.	It Returns the result.

3	used to Prevent function Recreation	used for expensive Calculations
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* Part 3:

Q11 What causes re-renders in React?

Ans A React component re-renders when the state changes (use state, useReducer), when props change, parent component re-renders and context value change. React re-renders to update the UI with the latest data.

Q12 Explain Reconciliation.

Ans Reconciliation is the process by which React updates the DOM efficiently.

→ Steps of Reconciliation.

1. React creates a new Virtual DOM.
2. Compares it with the Previous Virtual DOM.
3. Updates only the changed elements in the Real DOM.

Q13 What is Code Splitting in React?

Ans Code Splitting is a Technique to split your app into smaller bundles and load them only when needed. It is used with `React.lazy()`, `Suspense`.

Q14 What are Custom Hooks and Why useful?

Ans Custom Hooks are User-defined functions that use React Hooks to re-use logic. Custom Hooks are useful for

Better Separation Concern, to use Reusable logic, and write cleaner Code.

Q15 Difference between SPA and MPA

Ans	SPA	MPA
1.	It is Single Page Application.	It is Multipage Application.
2.	It loads one HTML page.	It loads Multiple Pages.
3.	SPA has faster Navigation	MPA has slower Navigation.
4.	It uses client-side routing ex! React Apps	It uses server-side routing. EX! Traditional Websites.